Find Permutation

You want to find a sequence $a := (a_1, a_2, \ldots, a_n)$ that is a permutation of $\{1, 2, \ldots, n\}$, subject to certain constraints.

You are given n, an integer m, and an array `c` containing m pairs. Each pair (i, j) in `c` constrains that $a_i < a_j$. Compute the unique a that satisfies all m constraints, if it exists.

Example Test Cases:

Test Case 1

Input: $n = 3$, $m = 2$, `c` = `[(1,2), (2,3)]`
Output: `[1, 2, 3]`
Explanation:
$a_1 < a_2$, since $1 < 2$, and $a_2 < a_3$, since $2 < 3$. This is the only permutation of $\{1, 2, 3\}$ satisfying the constraints.

Test Case 2

Input: $n = 3$, $m = 2$, `c` = `[(1,2), (1,3)]`
Output: `[]`
Explanation:
Both $(1, 2, 3)$ and $(1, 3, 2)$ are permutations of $\{1, 2, 3\}$ that satisfy the constraints, so a is not unique in this case.

Test Case 3

Input: $n = 6$, $m = 7$, `c` = `[(2,5), (5,1), (1,4), (4,3), (3,6), (2,1), (5,3)]`
Output: `[3, 1, 5, 4, 2, 6]`

Expected Time Complexity: $O(n + m)$

Input Format:
An integer n, an integer m, and an array `c` of length m.

Output Format:
Return an array of length n containing the unique sequence a satisfying all constraints. If such an a is not unique, return an empty array (the terminal output of your program will be `-1` in this case).

Constraints:

1. $2 \leq n \leq 2 \cdot 10^5$
2. $1 \leq m \leq 3 \cdot 10^5$
3. $1 \leq i, j \leq n$ for all (i, j) in `c`.
4. There is at least one sequence a that satisfies the constraints in `c`.

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the unique sequence, if it exists.

The `solve` function must compute this in $O(n + m)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.